

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4–Scenario Based Questions	Assessment:	Assessment Tool 1 – Scenario Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Excellent understanding and accurate application of population, sample, parameters, statistics, and point estimation.. (BL4)		
Student Name:	Kondeti . Lahari	Reg. No.:	192325118
Section / Batch:	B. Tech AIML	Date:	17-08-2026

Code of Conduct

I K. lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Lahari

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Q1. A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Given:

- University claims that at least 80% of students are satisfied.
- 95% Confidence Interval = (74%, 79%)

Hypotheses:

- $H_0: p \geq 0.80$
- $H_1: p < 0.80$

Analysis:

The 95% confidence interval for the satisfaction proportion is from 74% to 79%. The claimed value of 80% does not fall within this interval. In fact, the entire interval is below 80%.

Conclusion:

Therefore, the university's claim that at least 80% of students are satisfied with online classes is not supported by the survey data. The results indicate that the actual satisfaction proportion is likely to be less than 80%.

Q2. A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Given:

- Developer claims average usage = 30 minutes/day
- 99% Confidence Interval = (31.5, 35.2) minutes

Hypotheses:

- $H_0: \mu = 30$ minutes
- $H_1: \mu \neq 30$ minutes

Analysis:

The 99% confidence interval ranges from 31.5 to 35.2 minutes. The claimed average of 30 minutes is outside the confidence interval. Since the complete interval is greater than 30 minutes, the sample provides evidence that the actual average usage is higher than the claimed value.

Conclusion:

Therefore, the developer's claim that users spend an average of 30 minutes per day is not supported. The data suggests that users spend approximately 31.5 to 35.2 minutes per day on average.

Q3. A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Given:

- Company claims fraud detection rate = 95%
- p-value = 0.02
- Significance level $\alpha = 0.05$

Hypotheses:

- $H_0: p = 0.95$
- $H_1: p \neq 0.95$

Decision Rule:

If: $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$

Here:

$$0.02 < 0.05$$

Therefore, we reject H_0 .

Conclusion:

Since the p-value is less than the 5% significance level, there is statistically significant evidence against the company's claim that the fraud detection system identifies 95% of fraudulent transactions.

Q4. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given:

- Existing conversion rate = 12%
- New conversion rate = 15%
- p-value = 0.03
- $\alpha = 0.05$

Hypotheses:

- H_0 : New algorithm does not improve conversion rate.
- H_1 : New algorithm improves conversion rate.

Decision Rule:

If:

$$\text{p-value} < \alpha \rightarrow \text{Reject } H_0$$

Here:

$$0.03 < 0.05$$

Therefore, we reject H_0 .

Analysis:

The observed conversion rate increased from 12% to 15%, which is an improvement of 3 percentage points. The p-value of 0.03 indicates that this improvement is statistically significant at the 5% significance level.

Conclusion:

Therefore, the new recommendation algorithm significantly improves the conversion rate compared with the existing algorithm.

Q5. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given:

- Purpose: Determine whether the new system reduces waiting time.
- p-value = 0.08
- Significance level $\alpha = 0.05$

Hypotheses:

- H_0 : The new system does not reduce average waiting time.
- H_1 : The new system reduces average waiting time.

Decision Rule:

If:

$$\begin{aligned} \text{p-value} < \alpha &\rightarrow \\ &\text{Reject } H_0 \end{aligned}$$

If: Here:

$$\begin{aligned} \text{p-value} \geq \alpha &\rightarrow \\ &\text{Fail to reject} \\ &H_0 \end{aligned}$$

$$0.08 > 0.05$$

Therefore, we fail to reject H_0 .

Conclusion:

Since the p-value is greater than the 5% significance level, there is insufficient statistical evidence to conclude that the AI-based scheduling system reduces the average waiting time.

The hospital should not claim that the new system reduces waiting time based on this test.